

## CAP DE LA HEVE, France

WMO: 070280

Lat: 49.5089N

Lon: 0.0711E

Elev: 341

StdP: 14.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.1       | 29.7 | 18.3                          | 14.0 | 30.4 | 22.0 | 16.7 | 32.3 | 46.3                  | 48.4 | 41.9 | 48.4 | 10.1                  | 60   | 0.627 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 10.3                   | 80.9            | 66.9 | 76.7 | 64.9 | 73.1 | 63.3 | 68.3                | 78.0 | 66.3 | 73.6 | 64.8 | 70.8 | 8.7                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.9                            | 93.3 | 71.3 | 63.6 | 89.0 | 69.3 | 62.2 | 84.8 | 67.7 | 32.7          | 78.3 | 31.1 | 73.8 | 29.9 | 71.1 | 76.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| (4) 37.7          | 32.2 | 27.6 | 24.4                       | 88.3 | 3.2                | 4.5 | 22.1                                               | 91.5 | 20.3       | 94.2 | 18.5       | 96.7 | 16.1       | 100.0 |    |    |
| (5)               |      |      | 22.1                       | 71.6 | 3.3                | 2.2 | 19.7                                               | 73.1 | 17.7       | 74.4 | 15.9       | 75.6 | 13.4       | 77.2  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 52.9   | 42.1  | 42.7  | 46.3  | 50.3  | 55.3  | 60.6  | 64.2  | 64.3  | 60.9  | 55.6  | 48.7  | 43.6  |  |  |
|                                                                        | DBStd       | 9.58   | 6.04  | 5.48  | 5.18  | 5.45  | 5.51  | 5.18  | 4.65  | 4.72  | 4.52  | 5.19  | 5.00  | 6.05  |  |  |
|                                                                        | HDD50       | 962    | 247   | 208   | 137   | 59    | 11    | 0     | 0     | 0     | 1     | 14    | 80    | 205   |  |  |
|                                                                        | HDD65       | 4535   | 710   | 625   | 580   | 443   | 307   | 155   | 71    | 64    | 136   | 292   | 490   | 662   |  |  |
|                                                                        | CDD50       | 2037   | 2     | 4     | 22    | 67    | 175   | 317   | 439   | 445   | 329   | 189   | 40    | 8     |  |  |
|                                                                        | CDD65       | 133    | 0     | 0     | 0     | 1     | 6     | 22    | 44    | 44    | 14    | 2     | 0     | 0     |  |  |
|                                                                        | CDH74       | 682    | 0     | 0     | 0     | 2     | 24    | 155   | 223   | 221   | 53    | 4     | 0     | 0     |  |  |
|                                                                        | CDH80       | 164    | 0     | 0     | 0     | 0     | 1     | 37    | 58    | 59    | 9     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 12.0   | 14.1  | 13.5  | 12.5  | 11.2  | 10.9  | 10.0  | 10.2  | 10.2  | 10.5  | 13.2  | 13.8  | 14.4  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 33.5   | 2.9   | 2.4   | 2.1   | 2.2   | 2.3   | 2.5   | 2.2   | 2.7   | 2.8   | 3.7   | 3.8   | 3.9   |  |  |
|                                                                        | PrecMax     | 44.3   | 6.2   | 4.7   | 7.1   | 4.6   | 4.2   | 5.6   | 4.9   | 6.9   | 6.3   | 8.1   | 8.7   | 8.1   |  |  |
|                                                                        | PrecMin     | 20.9   | 0.3   | 0.5   | 0.4   | 0.2   | 0.5   | 0.4   | 0.4   | 0.6   | 0.4   | 0.9   | 0.9   | 0.9   |  |  |
|                                                                        | PrecStd     | 4.9    | 1.4   | 1.3   | 1.4   | 1.3   | 1.0   | 1.4   | 1.1   | 1.7   | 1.3   | 1.9   | 1.8   | 1.7   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 54.1  | 58.4  | 64.6  | 72.5  | 77.3  | 84.8  | 86.2  | 86.5  | 80.3  | 72.0  | 61.6  | 55.9  |  |  |
|                                                                        |             | MCWB   | 50.9  | 49.3  | 53.7  | 58.5  | 63.9  | 68.6  | 68.5  | 68.9  | 65.1  | 62.7  | 57.8  | 52.4  |  |  |
|                                                                        | 2%          | DB     | 52.2  | 53.4  | 60.0  | 66.4  | 72.4  | 78.7  | 79.8  | 80.1  | 73.7  | 67.3  | 58.4  | 54.2  |  |  |
|                                                                        |             | MCWB   | 49.9  | 48.4  | 51.7  | 55.7  | 60.9  | 66.5  | 66.8  | 66.1  | 63.1  | 60.9  | 55.1  | 51.7  |  |  |
|                                                                        | 5%          | DB     | 50.7  | 51.2  | 55.9  | 61.8  | 68.2  | 73.2  | 75.2  | 74.9  | 69.7  | 64.3  | 56.8  | 52.8  |  |  |
|                                                                        |             | MCWB   | 48.6  | 47.9  | 49.6  | 53.0  | 58.8  | 63.1  | 64.7  | 64.8  | 61.7  | 59.5  | 54.1  | 50.7  |  |  |
|                                                                        | 10%         | DB     | 49.2  | 49.4  | 53.1  | 58.0  | 64.2  | 68.7  | 71.4  | 70.9  | 67.1  | 62.2  | 55.4  | 51.3  |  |  |
|                                                                        |             | MCWB   | 47.3  | 46.9  | 48.8  | 51.4  | 57.2  | 61.1  | 63.3  | 63.3  | 60.9  | 58.4  | 53.0  | 49.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 52.1  | 51.3  | 55.3  | 60.1  | 65.7  | 70.2  | 71.1  | 70.8  | 67.4  | 63.8  | 58.8  | 53.9  |  |  |
|                                                                        |             | MCDB   | 53.1  | 54.7  | 62.2  | 68.3  | 75.4  | 82.4  | 83.0  | 82.3  | 75.0  | 69.3  | 60.8  | 55.0  |  |  |
|                                                                        | 2%          | WB     | 50.4  | 49.9  | 52.8  | 56.3  | 62.2  | 66.8  | 68.2  | 67.9  | 64.9  | 61.9  | 56.0  | 52.3  |  |  |
|                                                                        |             | MCDB   | 51.8  | 52.3  | 57.6  | 64.5  | 69.8  | 77.4  | 77.3  | 76.6  | 70.5  | 66.0  | 57.6  | 53.7  |  |  |
|                                                                        | 5%          | WB     | 49.0  | 48.5  | 51.0  | 54.2  | 59.8  | 64.1  | 66.0  | 65.9  | 63.3  | 60.3  | 54.7  | 51.0  |  |  |
|                                                                        |             | MCDB   | 50.4  | 50.4  | 54.6  | 60.6  | 66.3  | 72.0  | 72.3  | 72.1  | 67.9  | 63.4  | 56.4  | 52.5  |  |  |
|                                                                        | 10%         | WB     | 47.6  | 47.2  | 49.4  | 52.1  | 57.7  | 61.8  | 64.3  | 64.3  | 62.0  | 58.9  | 53.2  | 49.5  |  |  |
|                                                                        |             | MCDB   | 49.0  | 49.1  | 52.3  | 56.7  | 63.3  | 67.3  | 69.9  | 69.4  | 66.0  | 61.7  | 55.2  | 51.0  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.2   | 7.0   | 8.3   | 9.8   | 10.3  | 10.8  | 10.4  | 10.3  | 9.9   | 8.6   | 6.9   | 6.5   |  |  |
|                                                                        |             | MCDBR  | 6.3   | 9.4   | 13.2  | 15.6  | 17.0  | 19.2  | 18.5  | 17.0  | 14.6  | 10.7  | 8.0   | 6.7   |  |  |
|                                                                        |             | MCWBR  | 5.9   | 6.5   | 8.2   | 8.9   | 9.8   | 10.4  | 9.0   | 8.0   | 7.7   | 7.0   | 6.6   | 6.3   |  |  |
|                                                                        | 5% WB       | MCDBR  | 6.2   | 7.6   | 10.6  | 13.8  | 15.5  | 17.0  | 15.4  | 14.3  | 11.7  | 9.4   | 7.5   | 6.4   |  |  |
|                                                                        |             | MCWBR  | 6.2   | 6.2   | 7.3   | 8.4   | 9.5   | 10.1  | 8.4   | 8.4   | 7.3   | 7.0   | 6.9   | 6.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.326 | 0.343 | 0.377 | 0.395 | 0.400 | 0.400 | 0.394 | 0.389 | 0.373 | 0.358 | 0.333 | 0.320 |  |  |
|                                                                        | taud        |        | 2.433 | 2.410 | 2.315 | 2.285 | 2.314 | 2.347 | 2.364 | 2.399 | 2.427 | 2.457 | 2.453 | 2.402 |  |  |
|                                                                        | Ebn at Noon |        | 229   | 253   | 262   | 269   | 272   | 273   | 273   | 269   | 261   | 244   | 224   | 212   |  |  |
|                                                                        | Edh at Noon |        | 20    | 26    | 34    | 38    | 39    | 38    | 37    | 34    | 30    | 25    | 20    | 18    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 281   | 513   | 883   | 1381  | 1623  | 1811  | 1737  | 1459  | 1122  | 642   | 338   | 233   |  |  |
|                                                                        | RadStd      |        | 27    | 66    | 99    | 138   | 132   | 176   | 121   | 108   | 70    | 56    | 36    | 30    |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (45 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page